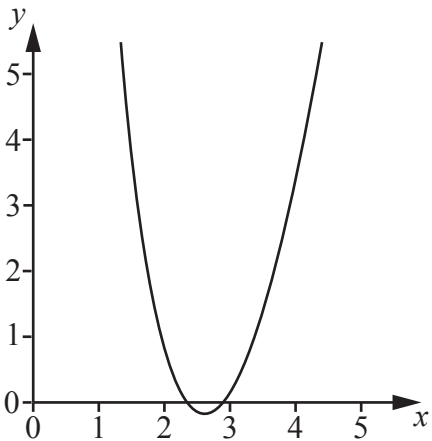
**6** The diagram shows the curve with equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = 13x - 34\ln(x) - 1.5$ for $x > 0$.




**(a)** A student attempts to locate a root of $\mathrm{f}(x) = 0$ by using a spreadsheet to calculate the values of $\mathrm{f}(x)$ for positive integer values of $x$.

Explain why the student's method may lead to the conclusion that $\mathrm{f}(x) = 0$ has no roots. **[1]**

**(b)** Determine the exact value of the $x$-coordinate of the turning point of the curve. **[2]**